

Object Oriented Programming

BCA — Semester 2 — — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

1. What is constructor? Explain its types with suitable example. How is the constructor called in a derived class in C++?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. What is Polymorphism in Object Oriented Programming? Implement a class Employee with a virtual function CalculateSalary() that returns the salary of the employee. Create two derived classes HourlyEmployee and SalariedEmployee. Override the CalculateSalary() function in both derived classes to calculate the salary of an hourly employee and a salaried employee respectively. Write a program that creates objects of both derived classes and calls the CalculateSalary() function for each.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What is operator overloading? Write a complete C++ program to overload + operator to add two objects of class 'Time'.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. What is the difference between public, private and protected inheritance in C++?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is exception handling? Explain types of exception handling and explain with suitable example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is friend function? Write a program to add private member of two different classes using friend function.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Write a program to demonstrate returning object from functions in C++.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. How 'this' pointer is used to resolve name conflict between local variable and member variable in C++? Explain with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Write a Program to read and write values through object using File Handling.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What do you mean by class template in C++? Write a program in C++ containing function template which determines the greater number between two integer inputs and two floating point inputs.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Differentiate between function overloading and function overriding in C++.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Illustrate the use of static variable with a simple program.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.